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RATIOS AND PROPORTIONS

4.1. What is a Ratio?

A ratio is a representation of distribution of a value present among the persons present and is shown as follows:

If a total is divided among A, B and C such that A got 4 parts, B got 5 parts and C got 6 parts then it is represented in ratio as $A:B:C = 4:5:6$.

So, $4:5:6$ means that the total value is divided into $4+5+6 = 15$ equal parts and then distributed as per the ratio.

Example 1: Divide Rs. 580 between A and B in the ratio of 14:15.

Solution: $A:B = 14:15 \Rightarrow 580$ is divided into 29 equal parts $\Rightarrow$ each part = Rs. 20.

So A's share = 14 parts = $14 \times 20 =$ Rs. 280

B's share = 15 parts = Rs. 300.

Example 2: If $A:B = 2:3$ and $B:C = 4:5$ then find $A:B:C$.

Solution: To combine two ratios the proportions common for them shall be in equal parts. Here the common proportion is B for the given ratios.

Making B equal in both ratios they become $8:12$ and $12:15 \Rightarrow A:B:C = 8:12:15$.

Example 3: Three numbers are in the ratio of 3 : 4 : 8 and the sum of these numbers is 975. Find the three numbers.

Solution: Let the numbers be $3x$, $4x$ and $8x$. Then their sum = $3x + 4x + 8x = 15x = 975 \Rightarrow x = 65$.

So the numbers are $3x = 195$, $4x = 260$ and $8x = 520$.

Example 4: Two numbers are in the ratio of 4 : 5. If the difference between these numbers is 24, then find the numbers.

Solution: Let the numbers be $4x$ and $5x$. Their difference = $5x - 4x = x = 24$ (given).

So the numbers are $4x = 96$ and $5x = 120$.

Example 5: Given two numbers are in the ratio of 3 : 4. If 8 is added to each of them, their ratio is changed to 5 : 6. Find two numbers.

Solution: Let the numbers be a and b .

$$A:B = 3:4 \Rightarrow \frac{A}{B} = \frac{3}{4}. \text{ Also, } \frac{(A+8)}{(B+8)} = \frac{5}{6}.$$

Solving we get, $A=12$ and $B = 16$.

